

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

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B.Tech (CSE/IT/ECE)/INTG(CSE/ECE), 1st Semester

Course Title: Software Development Fundamentals - I

Maximum Time: 2 hr.

Course Code: 15B11CI111

Maximum Marks: 35

Course Outcomes:

CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts.
CO2	Explain basics of C programming concepts to make decisions for solving problems.
CO3	Demonstrate and contrast different methods for writing modular programs in C.
CO4	Use various C programming constructs to implement iteration and recursion.
CO5	Apply and implement arrays, pointers, structures and file handling for solving real world problems.

Note: Attempt all questions

Q1. Draw a flowchart for finding all the roots of 'N' different quadratic equations (where N is provided by the user) of the form $ax^2+bx+c=0$, where a, b and c are real constants. Formula for calculating the roots of the quadratic equation is as follows:

$$\text{Roots} = \frac{-b \pm \sqrt{D}}{2a}$$

$$\text{where } D = (b * b) - (4 * a * c)$$

Your flowcharts should additionally show whether the roots will be "real and equal" or "real and unequal" or "imaginary".

[C01, 4 Marks, Understand]

Q2. Given an unsorted array **Arr** of N numbers (any combination of positive and negative). Your task is to create an array of alternate positive and negative numbers without changing the relative order of positive and negative numbers. It is ensured that the array has at least one positive number.

Note: Resultant array should start with a positive number and zero should be considered as a positive element.

Input: N = 9

Arr[] = {9,4,-2,-1,5,0,-5,-3,2}

Output: 9, -2, 4, -1, 5, -5, 0, -3, 2

Input: N = 10

Arr[] = {-5,-2,15,2,4,7,1,8,0,-8}

Output: 15, -5, 2, -2, 4, -8, 7, 1, 8, 0

[C02, 6 Marks, Understand]

Q3. Write a program to validate an IPv4 address. According to Wikipedia, IPv4 address are canonically represented in dot-decimal notation, which consists of four decimal numbers, each ranging from 0 to 255, separated by dots, i.e. 172.16.254.1

A valid IPv4 address is of the form $x_1.x_2.x_3.x_4$ where $0 \leq (x_1, x_2, x_3, x_4) \leq 255$. Thus we can write the generalized form of an IPv4 address (0-255). (0-255). (0-255). (0-255). Your task is to write a function **isValid()** which returns 1 if the given IPv4 address is valid, else return 0. The function takes the IPv4 address as the only argument in the form of string.

Note: Here we are considering numbers only from 0-255 and only additional leading zeros will be considered invalid. You are not supposed to use **strtoken()** function.

P.T.O

Sample Input: IPv4 address = 222.111.111.111
Output: 1
Sample Input: IPv4 address = 5555..555
Output: 0
Sample Input: IPv4 address = 02.111.111.111
Output: 0

[C03, 6 Marks, Understand]

Q4. Write a recursive function to print the Aliquot Sequence of the given number 'n'. Aliquot Sequence of a number start with itself, remaining terms of a sequence are sum of proper divisors of immediate previous term. For example, Aliquot Sequence for 10 is 10,8,7,1,0. The sequence may repeat. For example, for 6, we have an infinite sequence of all 6s. In such cases we print the repeating number and stop.

Explanation:

Input: n = 10

Output: 10 8 7 1 0

Sum of proper divisors of 10 is $5+2+1 = 8$

Sum of proper divisors of 8 is $4+2+1 = 7$

Sum of proper divisors of 7 is 1

Sum of proper divisors of 1 is 0

Note: There is no proper divisor of 1

Test Cases:

Input n: 6

Output = 6 repeats with 6

Input n: 12

Output = 12 16 15 9 4 3 1 0

[CO4, 6 Marks, Apply]

Q5. A testing agency is conducting an examination for class XIIth students of two different streams. Every student has name (string) and roll-no (integer). Additionally, a science stream's student has marks in physics, chemistry and mathematics while an arts student has the marks in English and regional language. The testing agency has decided to rank all the students (including both streams) as per their average marks. In case, there is a tie, then marks in physics for a science student and the marks in English for a arts student is given a preference to brake the tie. Write a C program that will take n students as input and display the details of top three students.

Note: Your program must ensure that there is a single structure created for storing the details of student from different streams. You are free to use additional data members to the structure if needed. Try to minimize the memory space used to store the examination information.

[C05, 7 Marks, Apply]

Q6. Indian mobile number consists of 10 digits. The first four digits from the left represent the area code while the next 6 digits represent the extension. For example, 9910235674 is a valid mobile number, where 9910 represent the area code, while 235674 represent the extension. A file called **MobileNumberRecord** contains 100 such mobile numbers written using the following structure:

```
struct MobileNo
{
    int area;
    int extension;
};
```

Write a C program that will create a new file called **Filtered** that stores all those mobile numbers which are divisible by the sum of digits of area code.

For example, 8034125730 is divisible by the sum of digits of its area code ($8+0+3+4=15$)

[CO4, 6 Marks, Apply]

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